

Component 1: User Registration

xPostForecast · Milestone 4: Component-Level Design · Pseudocode

Where it runs Backend: POST /auth/register (backend/routes/auth.js)

User stories US-1

Flowchart Component1_Registration_Flowchart.pdf

Description

Creates a new account from an email address, a username, and a password, after checking the input. Only a bcrypt hash of the password is ever stored.

Design principles

Information hiding. The plaintext password is hidden inside this component: it is hashed immediately, and only the hash is passed on to the users table. No other part of the system can see or leak the original password. The database details (connection settings, SQL) are likewise hidden behind UserStore, so this component only asks to insert a user.

Separation of concerns. Rate limiting is a separate concern handled by its own module (express-rate-limit) that runs before the registration logic, and duplicate detection is left to the database's UNIQUE constraints. The component's own code deals only with validating input and creating the account.

Pseudocode

```
BEGIN RegisterUser
  IF this address has made more than 20 attempts in the last 15 minutes THEN
    RESPOND 429 "Too many attempts. Please wait 15 minutes and try again."
    STOP
  END IF

  READ email, username, password FROM the request body

  IF email, username, or password is missing THEN
    RESPOND 400 "email, username, and password are required"
    STOP
  END IF

  IF the length of password is less than 8 THEN
    RESPOND 400 "Password must be at least 8 characters"
    STOP
  END IF

  SET passwordHash TO the bcrypt hash of password (work factor 10)
  INSERT username, email, passwordHash INTO the users table

  IF the insert failed THEN
    IF the failure was a duplicate username or email THEN
      RESPOND 409 "That username or email is already registered"
    ELSE
      LOG the error
      RESPOND 500 "Failed to register user"
```

```
END IF  
STOP  
END IF
```

```
RESPOND 201 "User registered successfully"  
END RegisterUser
```